



Walchand Institute of Technology, Solapur
(An Autonomous Institute)

Affiliated to
Punyashlok Ahilyadevi Holkar Solapur University,
Solapur

Choice Based Credit System (CBCS)

Structure and Syllabus
for
B.Tech. Computer Science and Engineering
W.E.F. 2026-27


Dean (Academic)


HEAD



Computer Science and Engineering

Vision

To develop professional engineers in Computer Science & Engineering having ethical values, research aptitude and ability to address challenges of modernization in the IT industry aiming at overall sustainable development of the society.

Mission

- M1 - To impart quality education in the field of Computer Science & Engineering in accordance with the needs of the Modernization & Globalization through technology enabled education.
- M2 - To inculcate lifelong learning in students to face challenges posed by ever-changing IT career landscape as a disciplined professional with a sense of professional ethics.
- M3 - To inculcate critical thinking and creativity for identifying various societal issues and to provide solutions.
- M4 - To enhance career opportunities for students through academia-industry interaction and research.

Computer Science and Engineering

Program Educational Objectives (PEOs)

1. Graduate will exhibit strong fundamental knowledge and technical skills in the field of Computer Science & Engineering to pursue successful professional career, higher studies and research.
2. Graduate will exhibit capabilities to understand and resolve various societal issues through their problem solving skills.
3. Graduate will be sensitive to ethical, societal and environmental issues as a software engineering professional and be committed to life-long learning.

Knowledge and Attitude Profile (WK)

WK1	A systematic, theory-based understanding of the natural sciences applicable to the discipline and awareness of relevant social sciences.
WK2	Conceptually-based mathematics, numerical analysis, data analysis, statistics and formal aspects of computer and information science to support detailed analysis and modelling applicable to the discipline.
WK3	A systematic, theory-based formulation of engineering fundamentals required in the engineering discipline.
WK4	Engineering specialist knowledge that provides theoretical frameworks and bodies of knowledge for the accepted practice areas in the engineering discipline; much is at the forefront of the discipline.
WK5	Knowledge, including efficient resource use, environmental impacts, whole-life cost, re-use of resources, net zero carbon, and similar concepts, that supports engineering design and operations in a practice area.
WK6	Knowledge of engineering practice (technology) in the practice areas in the engineering discipline.
WK7	Knowledge of the role of engineering in society and identified issues in engineering practice in the discipline, such as the professional responsibility of an engineer to public safety and sustainable development.
WK8	Engagement with selected knowledge in the current research literature of the discipline, awareness of the power of critical thinking and creative approaches to evaluate emerging issues.
WK9	Ethics, inclusive behavior and conduct. Knowledge of professional ethics, responsibilities, and norms of engineering practice. Awareness of the need for diversity by reason of ethnicity, gender, age, physical ability etc. with mutual understanding and respect, and of inclusive attitudes.

Program Outcomes (POs)	
PO 1	Engineering Knowledge: Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop to the solution of complex engineering problems.
PO 2	Problem Analysis: Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development. (WK1 to WK4)
PO 3	Design/Development of Solutions: Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture, society and environment as required. (WK5)
PO 4	Conduct Investigations of Complex Problems: Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis & interpretation of data to provide valid conclusions. (WK8).
PO 5	Engineering Tool Usage: Create, select and apply appropriate techniques, resources and modern engineering & IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems. (WK2 and WK6)
PO 6	The Engineer and The World: Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment. (WK1, WK5, and WK7).
PO 7	Ethics: Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national & international laws. (WK9)
PO 8	Individual and Collaborative Team work: Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams.
PO 9	Communication: Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations considering cultural, language, and learning difference
PO 10	Project Management and Finance: Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects and in multidisciplinary environments.

PO 11	Life-long Learning: Recognize the need for, and have the preparation and ability for i) independent and life-long learning ii) adaptability to new and emerging technologies and iii) critical thinking in the broadest context of technological change. (WK8)
Program Specific Outcomes (PSOs)	
1. Apply the principles of computational mathematics, computer systems and programming paradigms to solve computational problems. 2. Design and develop application software with functionalities applicable for desktop, web and mobile applications with due consideration of system software constraints. 3. Apply software engineering methods, cutting edge technologies and ICT, using appropriate tools and FOSS alternatives for designing ,developing& testing application software.	

Computer Science and Engineering Department

Legends Used

L	Lecture Hours / week
T	Tutorial Hours / week
P	Practical Hours / week
FA	Formative Assessment
SA	Summative Assessment
ESE	End Semester Examination
ISE	In Semester Evaluation
ICA	Internal Continuous Assessment
POE	Practical and Oral Exam
OE	Oral Exam
MOOC	Massive Open Online Course
HSS	Humanity and Social Science
NPTEL	National Programme on Technology Enhanced Learning
F.Y.	First Year
S.Y.	Second Year
T.Y.	Third Year
B. Tech.	Bachelor of Technology

Computer Science and Engineering									
Course Code Format									
2	1	I	T	U/P	2	C	C	1	T/L
Year of Syllabus revision		Program Code		U-Under Graduate P-Post Graduate	Semester No./ Year1/2/3/...8	CourseType		Course Serial No 1-9	T-Theory, L-Lab session P- Programming

Program Code	
CS	Computer Science and Engineering
Course Type	
BS	Basic Science
ES	Engineering Science
HU	Humanities & Social Science
MC	Mandatory Course
CC	Core Compulsory Course
SN*	Self-Learning <i>N* indicates the serial number of electives offered in the respective category</i>
EN*	Core Elective <i>N* indicates the serial number of electives offered in the respective category</i>
ON*	Open Elective <i>N* indicates the serial number of electives offered in the respective category</i>
SK	Skill Based Course
SM	Seminar
MP	Mini project
PR	Project
IN	Internship

Sample Course Code	
23CSU7CC1T	Information & Cyber Security

Computer Science and Engineering

B. Tech. Semester VII

Course Code	Name of Course	Engagement Hours			Credits	SA		FA		Total
		L	T	P		Theory	OE/POE	ISE	ICA	
23CSU7CC1T	Information & Cyber Security	3	--	--	3	60	--	40	--	100
23CSU7CC2T	Software Testing and Quality Assurance	3	--	--	3	60	--	40	--	100
23CSU7CC2A	Software Testing and Quality Assurance Tutorial	--	1	--	1	--	--	--	25	25
23CSU7EN*3T	Programme Elective Course IV	2	--	--	2	60	--	40	--	100
23CSU7EN*4T	Programme Elective Course V	2	--	--	2	60	--	40	--	100
23CSU7MD5T	Multidisciplinary Minor V : Software Testing and Quality Assurance	3	--	--	3	60	--	40	--	100
Subtotal		12	--		14	300	--	200	--	500
Laboratory Courses										
23CSU7CC1L	Information & Cyber Security	--	--	2	1	--	25	--	25	50
23CSU7EN*3L	Programme Elective Course IV	--	--	2	1	--	--	--	25	25
23CSU7EN*4L	Programme Elective Course V	--	--	2	1	--	--	--	25	25
23CSU7PR6L	Project Phase I	--	--	8	4	--	50	--	100	150
23CSU7MD5A	Multidisciplinary Minor V: Software Testing and Quality Assurance	--	1	--	1	--	--	--	25	25
Subtotal		--	2	14	8	--	75	--	225	300
Grand Total		12	2	14	22	300	75	200	225	800

Note:

- N*indicates the serial number of electives offered in the respective category
- ##indicates program code of offering Programme

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Computer Science and Engineering

B. Tech. Semester VIII

Course Code	Name of Course	Engagement Hours			Credits	SA		FA		Total
		L	T	P		Theory	POE	ISE	ICA	
23CMU8RN*1T	Research Methodology & IPR /MOOC#	2	--	--	2	50	--	--	--	50
23CSU8EN*2T	Programme Elective Course VI / MOOC#	2	--	--	2	50	--	--	--	50
Subtotal		4	--	--	4	100	--	--	--	100
Laboratory Courses										
23CSU8IN3T	Internship	--	--	20	10	--	50	--	100	150
23CSU8PR4L	Project Phase II	--	--	4	2	--	50	--	100	150
Subtotal		--	--	24	12	--	100	--	200	300
Grand Total		4	--	24	16	100	100	--	200	400

Note:

- N* indicates the serial number of electives offered in the respective category
- ## indicates program code of offering Program

List of Programme Elective-IV

Course Code	Name of Course
23CSU7E12T	Deep Learning
23CSU7E22T	Cloud Computing

List of Programme Elective-V

Course Code	Name of the Course
23CSU7E13T	Software Defined Networks
23CSU7E23T	Big Data Analytics

Note :

Program Elective VI' course offered by the department during Semester VIII in Self Learning

Course code	Remarks
23CSU8E12T	Edge Computing
23CSU8E22T	Blockchain and Its Applications
23CSU8E32T	Quantum Computing

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Note:

1. Batch size for the practical /tutorial shall be of 15 students. On forming the batches, if the strength of remaining students exceeds 7, then a new batch shall be formed.
2. Vocational Training (evaluated at Final Year B.Tech Semester VII) of minimum 15 days shall be completed in any vacation after S.Y. B.Tech Semester IV but before Final Year B.Tech Semester VII & the report shall be submitted and evaluated in Final Year B.Tech. Semester VII.
3. Appropriate Professional Elective IV Subjects may be added when required.
4. Project group for Final Year B.Tech. (Computer Science and Engineering) Semester VIII shall not be of more than five students.
5. Internship II / On Job Training (OJT):
Students may complete an internship / On Job Training (OJT) of a minimum of two months duration at the industry during Final Year Sem VIII.
6. The industry shall appoint a Supervisor to assess the performance of the student and share the same with the departmental supervisor for the fulfilment of ICA marks.
7. The student shall prepare a report of the work completed at the industry duly endorsed by the industry Supervisor and submit the same as an Internship report.
8. The ESE for Internship II / On Job Training (OJT) shall be conducted by the Departmental Supervisor in the presence of an external industry or academic expert.
9. **Internship should be of minimum 90 days.**
 - Internship may be offered by o Industry at their premises.
 - Industry at the institute campus.

Sr. No	Course Code	Option –I	Option II
1	23CMU8RN*1T	‘Research Methodology & IPR’ course offered by the department during Semester VIII in Self Learning / Online mode and successfully passing ESE for the same. 23CMU8R11T – Self Learning Mode	MOOC Certification* 23CMU8R21T – For MOOC (Option I) 23CMU8R31T – For MOOC (Option II)
2	23CSU8EN*2T	‘Program Elective VI’ course offered by the department during Semester VIII in Self Learning / Online mode and successfully passing ESE for the same. Program Elective VI options: 23CSU8E12T - Edge Computing 23CSU8E22T - Blockchain and Its Applications 23CSU8E32T - Quantum Computing	MOOC Certification* 23CSU8E42T – MOOC (Option I) 23CSU8E52T – MOOC (Option II) 23CSU8E62T – MOOC (Option III)

1. If students have not completed the MOOC in Semester V or VI, then students have to complete this MOOC in the Semester VII or VIII. In Semester VIII, the marks earned by the students in final assessment of this MOOC will be appropriately scaled and transferred for 23CMU8RN*1T and / or 23CSU8EN*2T along with the required credits.



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The list of approved MOOCs will be finalized and released by Board of Studies before the commencement of the semester as per the available courses on MOOC platform.

2. If students do not pass the MOOC examination of 23CMU8RN*1T or 23CSU8EN*2T before commencement of End Semester Examination (ESE) of Semester VIII, then they must appear for the institute-conducted ESE for courses (mentioned in option I) by department during Semester VIII.

• Multidisciplinary Minor (MDM) Courses

Sr. No.	MDM Program	MDM I (Sem III)	MDM II (Sem IV)	MDM III (Sem V)	MDM IV (Sem VI)	MDM V (Sem VII)
1	Mechanical and Automation Engineering	Manufacturing Processes and Mechanisms	Machine Drawing and 3D Modeling	Automotive Engineering and Robotics	Additive Manufacturing	Thermal Systems
2	Civil Engineering	Smart Buildings	Geoinformatics	Environmental Impact Assessment	Infrastructural Systems	Disaster Preparedness and planning
3	Electronics and Telecommunication Engineering	Fundamentals of Electronic Circuits	Electronics Design and Prototyping	Introduction to Embedded Systems	Fundamentals of Communication Techniques	Enclosure and Communication Design for IoT

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WALCHAND INSTITUTE OF TECHNOLOGY, SOLAPUR
(An Autonomous Institute)
B.Tech. (Computer Science and Engineering), Semester-VII

23CSU7CC1T – INFORMATION & CYBER SECURITY

Teaching Scheme		Examination Scheme	
Lectures	3 Hours/week	ESE	60 Marks
Practical	2 Hours/week	ISE	40 Marks
Credits	4	ICA	25 Marks
		OE	25 Marks

Course Objectives:

1. Explore the basic issues and foundational principles of computer and network security, including the various security attacks and services and illustrate the principles of modern symmetric and asymmetric ciphers.
2. To understand the knowledge required to implement security measures that enforce confidentiality, integrity, authentication, and non-repudiation in modern communication networks.
3. To Study the concepts of Message Authentication code and understand the issues related to Network Access Control and Cloud Security.

Course Outcomes:

At the end of this course, the student will be able to -

1. Describe the key security requirements, architecture for OSI, security types, threats and apply the concepts of symmetric cryptography.
2. Apply basic principles of public-key cryptosystems to provide security to the transmitted information and understand the cryptographic hash functions.
3. Explain concepts of Message Authentication code and understand the issues related to Network Access Control and Cloud Security.

Unit – I	Symmetric Ciphers	8 Hours
Computer Security Concepts, OSI Security Architecture, Attacks, Services, Mechanism, A model for Network security, Classical Encryption techniques – Symmetric Cipher model, Substitution Techniques, Transposition techniques.		
Unit – II	Block Cipher and Data Encryption Standard	8 Hours
Simplified DES, Block Cipher principles, The Data Encryption Standard, The strength of DES, Differential and Linear Cryptanalysis, Block Cipher design principles, Block Cipher Mode of Operation.		
Unit – III	Public Key Cryptography	7 Hours
Public Key Cryptography and RSA – Principles of Public Key Cryptosystems, The RSA Algorithm, Key management - Other public key cryptosystems – Key Management, Diffie-Hellman Key Exchange.		
Unit – IV	Cryptographic Hash Functions	8 Hours
Cryptographic Hash Functions: Applications of Cryptographic Hash Functions, Two Simple Hash Functions, Hash Functions Based on Cipher Block Chaining, Secure Hash Algorithm (SHA), SHA-3.		



Unit – V	Message Authentication Codes	7 Hours
Authentication requirements, Authentication Functions, Message Authentication Codes, Hash Functions, security of Hash Functions and MACS Digital Signatures. Authentication Protocols– Digital Signatures, Authentication Protocols, Digital Signature Standard.		
Unit – VI	Network Access Control	7 Hours
Network Access Control, Extensible Authentication Protocol, IEEE 802.1X Port-Based Network Access Control. Wireless Network Security: Wireless Security, Mobile Device Security, IEEE 802.11 Wireless LAN Overview, IEEE 802.11i Wireless LAN Security.		
Internal Continuous Assessment (ICA): 1. Write a program that can encrypt and decrypt using the Caesar cipher (Additive Cipher) 2. Write a program to encrypt and decrypt given text using the Affine Cipher 3. Write a program to encrypt and decrypt a message using Mono-alphabetic Cipher. 4. Write a program to encrypt and decrypt a message using Play fair Cipher. 5. Write a program to encrypt the message using the Hill cipher 6. Write a program to encrypt and decrypt a message using Poly alphabetic Cipher (Vigenere Cipher) 7. Write a program to encrypt and decrypt a message using One Time Pad (OTP)/Vernam Cipher 8. Implement Transposition Techniques Rail Fence Cipher 9. Write a program to simulate RSA algorithm 10. Write a program to demonstrate Diffie Hellman Key Exchange algorithm		
Text Books :		
1. Cryptography and Network Security - Principles and Practices, William Stallings Pearson Education Limited, 7th Edition, 2017. ISBN-13: 978-0134444284 ISBN-10:0134444280. 2. Cryptography and Network Security, Behrouz A. Forouzan, Tata McGraw-Hill, 2008, ISBN-13: 978-0-13-187319-3. RV College of Engineering® Digital Communication Engineering 10 3. Computer Security: Principles and Practice, William Stallings, Lawrie Brown, Pearson Education Limited, 4th Edition. ISBN-10: 9780134794105. 4. Cryptography and Network Security, Atul Kahate, Tata McGraw-Hill, 2003, ISBN-81:203-2186-3.		



WALCHAND INSTITUTE OF TECHNOLOGY, SOLAPUR
(An Autonomous Institute)
B.Tech. (Computer Science and Engineering), Semester-VII

23CSU7CC2T : SOFTWARE TESTING AND QUALITY ASSURANCE

Teaching Scheme		Examination Scheme	
Lectures	3 Hours/week	ESE	60 Marks
Tutorial	1 Hour/week	ISE	40 Marks
Credits	4	ICA	25 Marks

Course Objectives:

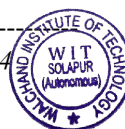
1. To describe knowledge of the software testing process and various methods of testing.
2. To describe generation and execution of test strategy & test plan.
3. To discover correctness, completeness and quality of software.
4. To implement automated testing tools to test software.

Course Outcomes:

At the end of this course, students will be able to

1. Compare the different software testing methods and select the suitable one for a given scenario.
2. Design test strategy & test plan for software testing.
3. Apply different approaches of quality assurance and standards for increasing quality of software.
4. Demonstrate automated testing tools to test software.

Unit – I	Fundamentals of Software Testing	8 Hours
Introduction, Basics of Software Testing, Approaches to Testing, Testing During Development Life Cycle, Essential of Software Testing, Features of Testing, Misconceptions About Testing, Principles of Software Testing, Waterfall Model and Agile Testing Fundamentals, Test Policy, Strategy, Planning, Process, Challenges in Testing, Test Team Approach, Methods, Defect Classification, Defect, Error, Mistake in Software, Defect Life Cycle.		
Unit – II	Methods of Testing	8 Hours
Software Verification and Validation, Black-Box and White-Box Testing, Static and Dynamic Testing, Black-Box Testing Techniques-Equivalence Partitioning, Data Testing, State Testing, Other Black Box Test Techniques. White-Box Testing Techniques-Data Coverage, Code Coverage, Other White Box Test Techniques.		
Unit – III	Levels of Testing	7 Hours
Verification and Validation Model, Levels of Testing, Proposal Testing, Requirement Testing, Design Testing, Unit Testing, Module Testing, Integration Testing, Big-Bang Testing, Sandwich Testing, System Testing- GUI Testing, Compatibility Testing, Security Testing, Performance Testing, Volume Testing, Stress Testing, Load Testing, Regression Testing, Smoke Testing, Sanity Testing, Ad hoc Testing, Usability Testing, Acceptance Testing-Alpha Testing, Beta Testing, Gamma Testing.		
Unit – IV	Test Planning & Documentation	8 Hours
Test Planning-The goal of Test Planning, Test Planning Topics, Writing and Tracking Test Cases-The Goal of Test Case Planning, Overview, Test Case Organization and Tracking, Bugs and Bug tracking.		



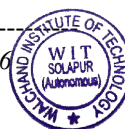
Unit – V	Quality Concepts & Software Quality Assurance	8 Hours
Quality Concepts-What is Quality?, Software Quality, The Software Quality Dilemma, Achieving Software Quality, Elements of Software Quality Assurance, SQA Processes and Product Characteristics, SQA Tasks, CMM, The SQA Plan.		
Unit – VI	Automated Testing and Testing Tools	6 Hours
Introduction, The Benefits of Automation and Tools, Software Test Automation, Random Testing, Realities of Using Test Tools and Automation.		
Internal Continuous Assessment (ICA): 1. Understand the requirement, generate the test case, analyse the bug using ChatGPT AI tool. 2. Develop a comprehensive test plan for a simple application (e.g., login system or calculator etc). 3. Use Claude AI for designing test strategy and complex test scenarios. 4. Understand and Apply black-box techniques to a sample problem. 5. Generate a basic automated test using Selenium Web Driver or Cypress. 6. Convert Manual test cases into automation script (Python / Java) using Copilot AI tool. 7. Use cursor AI tool and generate selenium for different scenarios. 8. Identify regression test cases for a feature update in an existing application using ChatGPT AI tool. 9. Perform API testing using POSTMAN tool. 10. Perform performance testing for a web application Using JMeter or LoadRunner.		
Text Books		
1. Software Testing Principles, Techniques and Tools By M G Limaye, Published by Tata McGraw-Hill Education Private Limited, Published 2009, ISBN (13): 978-0-07-013990-9, ISBN (10): 0-07-013990-3 (Chapter 1 & 3) 2. Software Testing, Second Edition By: Ron Patton, Published by SAMS, ISBN-13: 978-0672327988 ISBN-10: 0672327988 (Chapter 2, 4 & 6) 3. Software Engineering: A Practitioner Approach by Roger S Pressman, 8th Edition, Publisher McGraw Hill (Chapter 5)		
Reference Books		
1. Software Testing Principle and Practices By Ramesh Desikan, Gopalaswamy Ramesh, Pearson Education, ISBN 978-81-7758-121-8 2. Software Testing Principles and Practices By Naresh Chauhan, Publisher OXFORD UNIVERSITY PRESS-NEW DELHI, ISBN 0-19-806184-6 3. Beautiful Testing: Leading Professionals Reveal How They Improve Software By Adam Goucher, Tim Riley, Publisher O'reilly 4. Foundations of Software Testing By Rex Black, Dorothy Graham, Erik Van Veenendaal, Isabel Evans, Published by Cengage Learning India Pvt Ltd. 5. Lessons Learned in Software Testing by Cem Kaner, James Bach, Bret Pettichord, Publisher Wiley 6. Testing Computer Software Cem Kaner, Jack Falk, Hung Q. Nguyen, Publisher Wiley Selenium Testing Tools Cookbook By Unmesh Gundecha Published by Packt, ISBN: 978-1-84951-574-0 7. Dr. K.V.K.K. Prasad, "Software Testing Tools: Covering WinRunner, Silk Test, Load Runner, JMeter and TestDirector With Case Studies", Dreamtech Publications, ISBN: 10:81-7722-532-4		



WALCHAND INSTITUTE OF TECHNOLOGY, SOLAPUR
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B.Tech. (Computer Science and Engineering), Semester-VII

23CSU7E13T –Programme Elective IV: DEEP LEARNING

Teaching Scheme		Examination Scheme	
Lectures	2 Hours/week	ESE	60 Marks
Practical	2 Hours/week	ISE	40 Marks
Credits	3	ICA	25 Marks
Course Objectives:			
1. Understand the basics of neural networks and its need.			
2. Comparing different deep learning models.			
3. Understand the Convolutional and Recurrent nets, autoencoders and decoders in Deep Learning.			
Course Outcomes:			
Student will be able to-			
1. Understand the basics of Deep Learning and its applications.			
2. Apply the technique of Convolution (CNN) and Recurrent Neural Network (RNN) for Deep learning model development			
3. Design autoencoders and decoders for Deep Learning applications			
Unit – I	Foundations of Deep learning		7 Hours
What is machine learning and deep learning? Supervised and Unsupervised Learning, Limitations of machine learning, Advantages and challenges of deep learning. Understanding how deep learning works, Common Architectural Principles of Deep Network, Applications of Deep learning			
Unit – II	Deep Neural Networks (DNNs)		6 Hours
Introduction to Neural Networks: The Biological Neuron, The Perceptron, Multilayer Feed-Forward Networks, Training Neural Networks: Backpropagation and Forward propagation Activation Functions: Linear, Sigmoid, Tanh, Softmax, Rectified Linear, Loss Functions: Loss Function Notation, Loss Functions for Regression, Loss Functions for Classification, Gradient-Based Learning, Gradient Descent, vanishing and exploding gradient descent			
Unit – III	Convolution Neural Network (CNN)		5 Hours
Introduction, CNN architecture overview, The Basic Structure of a Convolution Network-Padding, Strides, Typical Settings, the ReLU layer, Pooling, Fully Connected Layer			
Unit – IV	Recurrent Neural Networks		7 Hours
Recurrent and Recursive Nets: Recurrent Neural Networks, Bidirectional RNN, Introduction to Long Short-Term Memory (LSTM), Gated Recurrent Units (GRUs), Applications of RNN in Real World-Image Captioning and Time Series Forecasting and Prediction			
Unit – V	Autoencoders		5 Hours
Under complete Auto encoders, Regularized Autoencoders-Sparse Autoencoders, Stochastic Encoders and Decoders, Applications of Autoencoders			



Internal Continuous Assessment (ICA):

ICA shall include at least eight of the following:

1. Implementation of a deep learning model with MNIST data using Caffe2
2. Implementation of CNN for flower Image classification
3. Build a MLP model to demonstrate the effect of Batch Normalization using Keras.
4. Implementation of CNN for Image classification using MNIST dataset
5. Implementation of CNN for Hand written digit recognition using MNIST dataset
6. Build a Keras classification Model on the Diabetes Dataset
7. Implementation of CNN model for animal image classification using Fashion MNIST
8. Implementation of sentiment analysis using LSTM using Amazon Alexa reviews Dataset.
9. Implementation of Airline sentiment analysis using LSTM using Tweeter reviews Dataset.
10. Implementation of Autoencoder techniques

Text Books

1. Goodfellow, I., Bengio, Y. Courville, A, “Deep Learning”, MIT Press, 2016.
2. Josh Patterson & Adam Gibson, “Deep Learning”
3. Charu Agarwal, “Neural Networks and deep learning”, A textbook
4. Deep Learning - A Practical Approach by Rajiv Chopra, Khana Publications, ISBN: 9789386173416

Reference Books

Josh Patterson, Adam Gibson "Deep Learning: A Practitioner's Approach", O'Reilly Media, 2017

e-Books:

1. <http://csis.pace.edu/ctappert/cs855-18fall/DeepLearningPractitionersApproach.pdf>
2. https://www.dkriesel.com/_media/science/neuronalenetze-en-zeta2-1col-dkrieselcom.pdf
3. MOOC Courses Links: <https://www.my-mooc.com/en/categorie/deep-learning>
4. Michael Nielsen, “Neural Networks and Deep Learning”, Online book, 2016
(<http://neuralnetworksanddeeplearning.com/>)
5. Deep Learning for Visual Computing https://onlinecourses.nptel.ac.in/noc22_ee54
6. Deep Learning - IIT Kharagpur https://onlinecourses.nptel.ac.in/noc22_cs22
7. Deep Learning - IIT Ropar https://onlinecourses.nptel.ac.in/noc22_cs35/
8. Introduction to Deep Learning:
<https://www.coursera.org/learn/introduction-to-deeplearning-boulder>
9. Deep Learning Specialization :
<https://www.coursera.org/specializations/deep-learning>



WALCHAND INSTITUTE OF TECHNOLOGY, SOLAPUR
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B.Tech. (Computer Science and Engineering), Semester-VII

23CSU7E23T – Programme Elective IV: CLOUD COMPUTING

Teaching Scheme		Examination Scheme	
Lectures	2 Hours/week	ESE	60 Marks
Practical	2 Hours/week	ISE	40 Marks
Credits	3	ICA	25 Marks

Course Objectives

1. Introduce students to the fundamental concepts, characteristics, and applications of cloud computing, including various cloud service and deployment models.
2. Enable students to understand and identify different cloud services and tools used for virtual machine provisioning, management, and migration to cloud environments.
3. Provide knowledge of hybrid cloud architectures by explaining the integration of public and private clouds, along with an understanding of cloud security challenges and protective measures.

Course Outcomes

At the end of this course, students will be able to

1. Understand the basics and applications of various cloud computing models.
2. Identify various services for VM provisioning and migration to cloud.
3. Discuss the integration of public and private cloud and cloud security measures.

UNIT I	Introduction to Cloud Computing	4 Hours
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History of Cloud Computing, Evolution of Cloud Computing, Traditional vs. Cloud Computing. Why Cloud Computing, Cloud service models (IaaS, PaaS & SaaS), Cloud deployment models (Public, Private, Hybrid and Community Cloud), Benefits and Challenges of Cloud Computing, Role of virtualization in Cloud Computing.

UNIT II	Virtual Machines Provisioning and Migration Services	5 Hours
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Virtual Machines Provisioning and Manageability, Virtual Machine Migration Services, VM Provisioning and Migration in Action, Provisioning in the Cloud Context.

Unit III	Understanding Services and Applications by Type	6 Hours
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Defining Infrastructure as a Service (IaaS), IaaS workloads, Pods, aggregation and silos, Defining Platform as a Service (PaaS), Defining Software as a Service (SaaS), SaaS characteristics, Open SaaS and SOA, Salesforce.com and CRM SaaS, Defining Identity as a Service (IDaaS), What is an identity?, Networked identity service classes, Identity system codes of conduct, IDaaS interoperability, User authentication, Authorization markup languages, Defining Compliance as a Service (CaaS).



Unit IV	Integration of Private and Public Clouds	6 Hours
Aneka: Introduction, Technologies and Tools for Cloud Computing, Aneka Cloud Platform, Aneka Resource Provisioning Service, Hybrid Cloud Implementation, CometCloud: Architecture, Autonomic Behavior of CometCloud, Overview of CometCloud-based Applications.		
Unit V	Understanding Cloud Security	5 Hours
Securing the Cloud, the security boundary, security service boundary, security mapping, Securing Data, brokered cloud storage access, storage location and tenancy, encryption, auditing and compliance, Establishing Identity and Presence, identity protocol standards, Windows Azure identity standards, Presence.		
Unit VI	Migration to Cloud	4 Hours
When and not to migrate to Cloud, migration paths for cloud, selection criteria for cloud deployment, issues/risks in migrating to cloud computing.		
Internal Continuous Assessment (ICA): Minimum 8 assignments similar those in the following list, which require students to identify and implement the use case scenarios for Cloud and Cloud enabled technologies mentioned above: - To identify the limitations of the traditional on-premises model for a school management system and propose a cloud-based solution using AWS. - To estimate the monthly cost of hosting a simple web application on AWS. - To design a deployment strategy for a global news website. - To secure an AWS account for a startup. - To create a secure network architecture for a web application. - To deploy a basic web application using AWS compute services. - To design storage solution for an application with different data needs. - To implement a database solution for an online bookstore. - To design a highly available and scalable cloud architecture. - To ensure high performance for a traffic-intensive application.		
Text Books		
- Cloud Computing: Principles and Paradigms by Rajkumar Buyya, James Broberg, Andrzej Goscinski, John Wiley and Sons Inc. - Cloud Computing Bible by Barrie Sosinsky, Wiley Publishing, Inc. https://arpitapatel.files.wordpress.com/2014/10/cloud-computing-bible1.pdf - Cloud Computing for Dummies by Judith Hurwitz, Marcia Kaufman, Fern Halper, Robin Bloor, John Wiley and Sons Inc. - Cloud Computing Black Book by Kailash Jayaswal, Jagannath Kallakurchi, Donald J. Houde, Dr. Deven Shah, Kogent Learning Solutions Inc.		
Reference Books		
- Cloud Security: A Comprehensive Guide to Secure Cloud Computing by Ronald L. Krutz and Russell Dean Vines, Wiley Publishing Inc. - Cloud Computing: Implementation, Management and Security by John W. Rittinghouse, James F. Ransome, CRC Press Taylor & Francis Group. - Mastering Cloud Computing by Rajkumar Buyya, Christian Vecchiola, S. Thamarai Selvi, Tata McGraw-Hill publications. - Cloud Computing: A Practical Approach by Anthony T. Velte, Toby J. Velte and Robert Elsenpeter, The McGraw-Hill Companies.		



WALCHAND INSTITUTE OF TECHNOLOGY, SOLAPUR
(An Autonomous Institute)
B.Tech. (Computer Science and Engineering), Semester-VII

23CSU7E14T - Programme Elective-V: Software Defined Networks

Teaching Scheme		Examination Scheme	
Lectures	2 Hours/week	ESE	60 Marks
Practical	2 Hours/week	ISE	40 Marks
Credits	3	ICA	25 Marks

Course Objectives:

1. To develop students' ability to understand the concepts of traditional networks with its limitations and the need to move to Software Defined Networks
2. To develop the students' ability to understand the fundamentals of SDN, its planar architecture and to understand the flexibility of multilevel pipeline processing.
3. To provide the students with knowledge of the working of SDN between the controller and data plane and emphasis on the table matching.
4. To create switches and designing networks by manually adding/deleting flow entries inside the table and learning to dissect the packets.

Course Outcomes:

1. At the end of the course, students will be able to
2. Understand the design principles and performance enhancement strategies that adopted in performance evolution of different network components.
3. Solve the performance related problems of SDN, including those in routing, optimizing traffic engineering.
4. Analyze the performance of routing, optimizing traffic engineering using SDN.

Unit I	Introduction to SDN	6 Hours
• Traditional Networks: Data, Control, and Management planes, limitations, and rigid architecture. • SDN Evolution: Genesis, need for SDN, and foundational characteristics. • SDN Architecture: Data plane, control plane, and application plane. • Centralized vs. Distributed Control: Pros and cons.		
Unit- II	OpenFlow and Data Plane	6 Hours
• OpenFlow Protocol: Specification 1.0–1.3, flow tables, flow table pipeline processing, and table matching. • Data Plane Operations: Forwarding rules, group tables, and meter tables. • Southbound Interface: Interaction between switches and controllers.		
Unit III	SDN Controllers and Northbound API	6 Hours
• Control Plane Functions: Network Operating Systems (NOS), controller implementations (POX, NOX, Ryu, Open Daylight, ONOS). • Northbound API: Programming languages, Application Programming Interfaces (APIs), and Network Application development.		



Unit IV	Network Function Virtualization (NFV)	6 Hours
• NFV Concepts: Benefits, requirements, and reference architecture. • Virtualization: Virtual LANs (VLANs), OpenFlow support, and network slicing. • NFV Infrastructure: Virtualized Network Functions (VNFs), management, and orchestration.		
Unit V	SDN Applications and Simulation	6 Hours
• SDN Use Cases: Data center networking, WAN, and ISP automation. • Tools: Mininet emulator, building custom topologies, and packet dissection via Wireshark. • Security: Security as a service using SDN/NFV.		
Internal Continuous Assessment (ICA): ICA shall consist of minimum eight experiments based on below content. 1. Mininet Setup: Creating custom topologies and testing connectivity 2. OpenFlow Switch Configuration: understanding OpenFlow-enabled switch behaviour 3. Flow Table Inspection and Management: add, delete and display OpenFlow entries 4. Packet Analysis: Viewing OpenFlow messages in Wireshark 5. Controller Installation: Configuring ONOS or OpenDaylight 6. Programming: Implement Static Routing Using SDN controllers 7. Programming: Implementing firewall rules using Python/SDN controllers 8. VLAN Configuration in SDN: Configure VLANs, verify traffic isolation, analyse forwarding behaviour. 9. SDN-Based Load Balancer: implement basic load balancing logic using OpenFlow 10. SDN-Based Traffic Engineering: Optimize traffic by configuring alternate forwarding paths, observe load distribution, analyse routing performance.		
Text Books		
1. Nadeau, Thomas D., and Ken Gray. SDN: Software Defined Networks: an authoritative review of network programmability technologies. " O'Reilly Media, Inc.", 2013. 2. Chuck Black and Paul Goransson, "Software Defined Networks: A Comprehensive Approach", Morgan Kaufman. 3. Coker, Oswald, and Siamak Azodolmolky. Software-defined Networking with OpenFlow: Deliver Innovative Business Solutions. Packt Publishing Ltd, 2017. 4. Software Defined Networks: Architecture and Applications by Anand Nayyar, Bhawna Singla, Preeti Nagrath, Willey Publication		
e-sources		
1. https://www.opennetworking.org/wp-content/uploads/2014/10/openflow-spec-v1.3.0.pdf (Open Flow version 1.3) 2. http://mininet.org/ (Mininet Network Emulator) 3. https://www.opennetworking.org/ (Open Network Foundation)		



WALCHAND INSTITUTE OF TECHNOLOGY, SOLAPUR
(An Autonomous Institute)
B.Tech. (Computer Science and Engineering), Semester-VII

23CSU7E24T - Programme Elective V: BIG DATA ANALYTICS

Teaching Scheme		Examination Scheme	
Lectures	2 Hours/week	ESE	60 Marks
Practical	2 Hours/week	ISE	40 Marks
Credits	3	ICA	25 Marks

Course Objectives

1. Understand Big Data fundamentals by identifying limitations of traditional DBMS and the need for Big Data Analytics.
2. Explain and apply Hadoop ecosystem components including HDFS, MapReduce, and YARN for large-scale distributed data processing.
3. Explore and utilize NoSQL databases such as MongoDB and Cassandra for managing unstructured and semi-structured data.
4. Apply Hive and Pig tools for querying, ETL processing, and analytics on big data platforms.

Course Outcomes

At the end of this course, students will be able to

1. Comprehend limitations of conventional DBMS and recognize need for Big Data Analytics.
2. Analyze the architecture, components, and features of Hadoop, and apply HDFS, MapReduce, and YARN to process and manage large-scale distributed data effectively.
3. Apply NoSQL database concepts by utilizing MongoDB and Cassandra features.
4. Use Hive and Pig to process and analyze big data on Hadoop using HQL, Pig Latin, and ETL techniques.

Unit I	Introduction to Types of Digital Data	3 Hours
Classification of Digital Data, Structured Data, Sources of structured data, Ease with Structured data, Semi-Structured data, sources of semi-structured data, Unstructured data, sources of unstructured data, Issues with terminology, Dealing with unstructured data.		
Unit II	Introduction to Big Data	3 Hours
Big data, What is big data? Why big data?, Other characteristics of data which are not definitional traits of big data, Challenges with big data, Big data stack		
Unit III	Big Data Analytics	4 Hours
Big Data Analytics, Analytics 1.0, Analytics 2.0, Analytics 3.0, Traditional BI vs. Big Data Environment, Terminologies used in Big Data Environment, Big Data Technology Landscape, NoSQL Databases, NoSQL Vs. RDBMS, NewSQL, Hadoop, Hadoop 1.0 vs. Hadoop 2.0, Data Science and Data Scientist.		



Unit IV	Introduction to Hadoop	5 Hours
Introducing Hadoop, Why not RDBMS, Distributed Computing Challenges, A Brief History of Hadoop, Hadoop Overview, Hadoop Components, High Level Architecture of Hadoop, Hadoop Distributed File System, HDFS Architecture, Daemons Related to HDFS, Working with HDFS Command, Special Features of Hadoop, Processing Data With Hadoop, Introduction How Map Reduce Works, Map Reduce Example, Word Count Example using Java Managing Resources and Applications with YARN Introduction, Limitation of Hadoop 1.0, Hadoop 2: HDFS, Hadoop 2: YARN, Interacting with Hadoop EcoSystem Hive, Pig, HBase, Sqoop.		
Unit V	Introduction to MongoDB	4 Hours
Recap of NoSQL databases, MongoDB – CRUD, MongoDB- Arrays, Java Scripts, Cursors, Map Reduce Programming, Aggregations.		
Unit VI	Introduction to Cassandra	3 Hours
Features of Cassandra, CQLSH - CRUD, Collections, Counter, List, Set, Map, Tracing.		
Unit VII	Introduction to Hive	4 Hours
What is Hive? History of Hive and Recent Releases of Hive, Hive Features, Hive Integration and Work Flow, Hive Data Units, Hive Architecture, Hive Primitive and Collection Data Types, Hive File Format, Hive Query Language (HQL) – Statements – DDL, DML. Hive Partitions – Bucketing, Views, Sub Query, Joins, Hive User Defined Function, Aggregations in Hive, Group by and Having, Serialization and Deserialization, Hive Analytic Functions.		
Unit VIII	Introduction to Pig	4 Hours
Introducing Pig, History and Anatomy of Pig, Pig on Hadoop, Pig Philosophy, ETL Processing, Pig Latin Overview, Word count example using Pig.		
Internal Continuous Assessment (ICA) : - Objective of assignments should be to test students understanding and assess their ability to put into practice the concepts and terminologies learned. - Assignments must be of nature, which require students to identify the use case scenarios for using technologies mentioned in syllabus. - It should consist of the 08-10 practical based on following guidelines. - Basic big data operations using NumPy, SciPy & Pandas. - Implementation of Plotting, Filtering and Cleaning a CSV File Data Using NumPy & Pandas. - Simple MongoDB and its CRUD Operations - Performing import, export and aggregation in MongoDB. - Performing CRUD operations using Cassandra. - Store the login details of the user such as UserID and Password. The information stored should expire in a day's time using time to live (TTL). - Map-Reduce Programming examples - Partitioning and processing using Hive. - Perform group by, order by, sort by, cluster by, distribute by queries using Hive. - Find out frequency of each word (word count) using pig.		

Text Book
1. Big Data and Analytics, Seema Acharya, Subhashini Chellappan, - Wiley India Pvt. Ltd. 2. Hadoop: The Definitive Guide, 3rd Edition, Tom White , - O'reilly Media. 3. Programming Hive, Edward Rutherglen, Dean Wampler, Jason Rutherglen, EdwardCapriolo. - O'reilly Media. 4. The Definitive Guide to MongoDB: A Complete Guide to Dealing with Big Data Using MongoDB (Definitive Guide Apress) 2e by David Hows, Eelco Plugge, Peter Membrey, Tim Hawkins. 5. Programming Pig, by Alan Gates - O'reilly Media. 6. Cassandra: The Definitive Guide, Eben Hewitt - O'reilly Media.
Reference Book
1. Big Data for Dummies, Judith Hurwitz, Alan Nugent, Dr. Fern Halper, Marcia Kaufman, Wiley Brand. 2. Big Data, Big Analytics: Emerging Business Intelligence and Analytic Trends for Today's Businesses (Wiley CIO), Michael Minelli, Michele Chambers, AmbigaDhiraj : John Wiley & Sons. 3. Mining of Massive Datasets, Anand Rajaraman, Jure Leskovec, Jeff rey D. Ullman, Cambridge University Press. 4. Hadoop in Action, Chuck Lam, Dreamtech Press, ISBN : 978-81-7722-813-7.

**WALCHAND INSTITUTE OF TECHNOLOGY, SOLAPUR****(An Autonomous Institute)****Final Year B.Tech. (Computer Science and Engineering), Semester-VII****23CSU7PR6L: Project Phase-I**

Teaching Scheme		Examination Scheme	
Lectures	--	ICA	100 Marks
Practical	8 Hours/week	POE	50 Marks
Credits	4		
Introduction:			
Project-based learning is a well-established educational paradigm gaining increasing importance today. To align with this approach, a project course is integrated into the final year curriculum, spanning both semesters. In this course, students work in teams to undertake a project, allowing them to showcase their abilities and develop expertise in their chosen areas of interest. The projects can involve both hardware and software, with a strong emphasis on design and research aspects. Additionally, effective communication, both oral and written, is a crucial skill for engineering graduates in various contexts. This course aims to cultivate these essential communication skills as well.			
Course Prerequisite:			
Basic knowledge of all the core subjects completed during the previous semesters.			
Course Objectives:			
1. To expose students to software engineering industries, enabling them to identify problem areas and formulate problem statements. 2. To equip students with the ability to design electronic and software systems that addresses the identified problems. 3. To develop students' skills in designing the hardware and software architecture for their projects. 4. To train students in writing technical specifications and project documentation for their chosen problems.			
Course Outcomes:			
After completing the course, students will be able to 1. Study and select problem of societal relevance. 2. Select an appropriate solution design with due consideration for society. 3. Carry out impact analysis for environment and sustainability consideration(s). 4. Design a system using software engineering techniques and modern tools. 5. Engage in team work and communicate effectively, while observing professional ethics. 6. Inculcate habit of self-study and lifelong learning.			



Guidelines for Project content:

Project Termwork: The termwork under project submitted by students shall include:

a. Work diary and weekly reporting

Work diary maintained by group and countersigned by the guide weekly. The contents of work diary shall reflect the efforts taken by project group for:

1. Searching suitable project work.
2. Brief report, preferably on Journals/conference papers/books or literature surveyed to select and bring out the project.
3. Brief report of feasibility studies carried to implement the project objectives.
4. Proposed diagram/design calculations, etc.

b. Synopsis:

The group should submit the synopsis (of 4-5 pages) in following form.

1. Title of Project
2. Names of Students
3. Name of Guide
4. Proposed work (must indicate the scope of the work and weekly plan up to march end)
5. Approximate expenditure (if any).

The synopsis of project is expected to approve by the guide and endorsed by the Head of the Department.

Note:- The project group consist of four students. The group has to submit the report on the progress of project work at the end of first semester.

		WALCHAND INSTITUTE OF TECHNOLOGY, SOLAPUR (An Autonomous Institute) B.Tech. (Computer Science and Engineering), Semester-VIII	
24CMU8R11T - RESEARCH METHODOLOGY AND IPR			
Teaching Scheme		Examination Scheme	
Lectures	2 Hours/week	ESE	50 Marks
Credits	2		
Course Prerequisite			
Basic knowledge of computer networks and cloud computing.			
Course Objectives:			
1. To introduce students to the principles, and processes of research. 2. To enable students to review literature and maintain it. 3. To introduce the different research design, data collection methods and analysis. 4. To provide knowledge of IPR and their role in innovation. 5. To describe the process for filing patent.			
Course Outcomes:			
At the end of this course, students will be able to 1. Comprehend the knowledge on basics of research and its types and Conduct a systematic literature review 2. Apply systematic methods to identify and define a research problem, and implement appropriate research and experimental design principles to develop an effective research framework 3. Employ suitable data collection strategies and statistical analysis techniques. 4. Demonstrate the concepts of Intellectual Property Rights in engineering.			
Unit – I	Introduction to Research Methodology		4 Hours
Objectives of Research, Motivation in Research, Types of Research, Research Approaches, Significance of Research, Research Methods versus Methodology, Research and Scientific Method, Importance of Knowing How Research is Done, Research Process, Criteria of Good Research			
Unit – II	Literature Survey		4 Hours
Literature review, Sources of literature, Various referencing styles, Maintain literature data using Mendeley, Identifying the research areas from the literature review and research database, Indexing and Abstracting.			
Unit – III	Defining the Research Problem and Research Design		7 Hours
Defining the Research Problem - What is a Research Problem?, Selecting the Problem, Necessity of Defining the Problem, and Technique Involved in Defining a Problem. Research Design - Important Concepts Relating to Research Design, Different Research Designs, Basic Principles of Experimental Designs			

Unit – IV	Data Collection & Analysis	7 Hours
Collection of Primary, secondary and other data collection methods, Processing and Analysis of data - Processing Operations, Elements/Types of Analysis, Statistics in Research, Simple Regression Analysis, Multiple Correlation and Regression, Partial Correlation.		
Unit – V	Introduction to Intellectual Property Rights (IPR)	4 Hours
Concept and need for IPR. Characteristics of IP, Types of IPR: Patents, copyrights, trademarks, trade secrets, industrial designs, geographical indications.		
Unit – VI	Patents Searching and Filing	4 Hours
Patent Information System, Patent Search & Patent Databases, Various types of Searches using Patent document, Elements of Patentability, Procedure for Filing of Patent Application and Types of Applications		
Text Books		
1. C.R. Kothari – Research Methodology: Methods and Techniques (New Age International Publishers, PDF available online). 2. Vinayak Bairagi, Mousami V. Munot, Vinayak Bairagi, Mousami V. Munot, Research Methodology, A Practical and Scientific Approach, Publisher: Chapman and Hall/CRC ,Year 2019, ISBN 9781351013253 3. Kumar, R. (2018). Research methodology: A step-by-step guide for beginners. 4. Intellectual Property Rights – Law & Practice		
Reference Books		
1. Donald R. Cooper & Pamela S. Schindler – Business Research Methods 2. John W. Creswell – Research Design: Qualitative, Quantitative, and Mixed Methods 3. Deborah E. Bouchoux – Intellectual Property: The Law of Trademarks, Copyrights, Patents 4. P. Narayanan – Intellectual Property Law 5. Prabuddha Ganguli – Intellectual Property Rights: Unleashing Knowledge Economy 6. T. Ramappa – Intellectual Property Rights under WTO 7. Uma Sekaran – Research Methods for Business (Open access versions available) 8. WIPO – Intellectual Property Handbook (World Intellectual Property Organization, free PDF). 9. OECD – Intellectual Property Rights and Innovation (Open access reports).		



WALCHAND INSTITUTE OF TECHNOLOGY, SOLAPUR
(An Autonomous Institute)
B.Tech. (Computer Science and Engineering), Semester-VIII

23CSU8EN12T - Programme Elective VI – EDGE COMPUTING

Teaching Scheme		Examination Scheme	
Lectures	2 Hours/week	ESE	50 Marks
Credits	2		
Introduction			
This course introduces edge computing fundamentals and explains how it overcomes cloud limitations like latency and bandwidth in real-time applications. It covers edge-cloud frameworks, core technologies, distributed data analytics, and edge service platforms. The course also explores AI, machine learning, and deep learning use cases such as autonomous vehicles and predictive maintenance.			
Course Prerequisite			
Basic knowledge of computer networks and cloud computing.			
Course Objectives:			
1. Develop foundational understanding of edge computing, its evolution from cloud computing, and its necessity for supporting low-latency and real-time applications. 2. Introduce key architectures, frameworks, and distributed system concepts required for designing and managing edge-based computing environments. 3. Provide hands-on insight into core edge technologies including Docker, Kubernetes, MQTT, Kafka, key-value stores, and edge data centre services. 4. Explore machine learning and deep learning techniques for on-device inference and demonstrate how AI models enable intelligent edge applications such as predictive maintenance and autonomous systems.			
Course Outcomes:			
At the end of course the student will be able to 1. Explain the principles, limitations, and architectures of edge computing and evaluate its role in addressing cloud latency and bandwidth constraints. 2. Apply distributed computing concepts-such as time synchronization, RTT, and distributed snapshots-to analyze edge system behavior. 3. Construct and manage edge pipelines using containerization (Docker, Kubernetes) and messaging frameworks (MQTT, Kafka) for real-time distributed analytics. 4. Implement and assess machine learning and deep learning models for edge-based sensor processing, predictive maintenance, and latency-critical applications.			
Unit – I	Evolution from Cloud to Edge Computing		3 Hours
Introduction to Cloud and its limitations to support low latency and RTT. From Cloud to Edge computing: Waves of innovation, Introduction to edge computing, edge computing paradigms.			
Unit – II	Foundations and Architectures of Edge Computing		4 Hours
Introduction to Edge Computing Architectures, Three Tier Architecture, Edge-only, Thick Edge + Cloud, Thin/Micro Edge + Cloud.			



Unit – III	Distributed Systems Concepts for Edge Environments	4 Hours
Edge Computing to support User Applications (5G-Slicing, self-driving cars and more)		
Unit – IV	Distributed Systems in Edge Computing	4 Hours
Concepts of distributed systems in edge computing such as time ordering and clocksynchronization, distributed snapshot, etc.		
Unit – V	Edge Data Centres and Lightweight Edge Cloud Services	4 Hours
Introduction to Edge Data Centre, Lightweight Edge Clouds and its services provided by different service providers.		
Unit – VI	Containerization and Orchestration for the Edge Ecosystem	3 Hours
Introduction to docker container and Kubernetes in edge computing. Design of edge storage systems like key-value stores.		
Unit – VII	Edge Messaging Pipelines and Analytics for IoT, M2M, and WSN	4 Hours
Introduction to MQTT and Kafka for end-to-end edge pipeline. Edge analytics topologies for M2M and WSN network (MQTT)		
Unit – VIII	Machine Learning and On-Device Deep Learning for Edge Applications	4 Hours
Use cases of machine learning for edge sensor data in predictive maintenance, image classifier and self-driving cars. Deep Learning On-Device inference at the edge to support latency-based application		
Text Books		
1. Fog and Edge Computing: Principles and Paradigms, Rajkumar Buyya (Editor), Satish Narayana Srirama (Editor), Wiley, 2019 2. Cloud Computing: Principles and Paradigms, Editors: Rajkumar Buyya, James Broberg, Andrzej M. Goscinski, Wiley, 2011 3. Cloud and Distributed Computing: Algorithms and Systems, Rajiv Misra, Yashwant Patel, Wiley 2020.		
e-Resources:		
Edge Computing - IIT Kanpur https://nptel.ac.in/courses/106104449		



WALCHAND INSTITUTE OF TECHNOLOGY, SOLAPUR
(An Autonomous Institute)
B.Tech. (Computer Science and Engineering), Semester-VIII

23CSU8EN22T - Programme Elective VI – BLOCKCHAIN and ITS APPLICATIONS

Teaching Scheme		Examination Scheme	
Lectures	2 Hours/week	ESE	50 Marks
Credits	2		
Introduction			
This course will cover the basic design principles of Blockchain technology and its application over different sectors, along with permissionless blockchain platforms - Ethereum, and one permissioned blockchain platform - Hyperledger.			
Course Prerequisite			
An individual should have a good knowledge of distributed systems, networking, cryptography, data structures, basic knowledge of Linux commands.			
Course Objectives:			
1. Understand how blockchain systems (Bitcoin and Ethereum) work, 2. Design, build, and deploy smart contracts and distributed applications, 3. Integrate ideas from blockchain technology into their own projects.			
Course Outcomes:			
At the end of course the student will be able to 1. Interpret basics of Blockchain Technology. 2. Differentiate between various consensus mechanisms 3. Design, build, and deploy a smart contract and distributed application.			
Unit – I	Introduction to Blockchain Technology		6 Hours
What is Blockchain , Basic Cryptographic Primitives – I , Basic Cryptographic Primitives – II, Basic Cryptographic Primitives – III, Basic Cryptographic Primitives – V , Distributed Systems for Decentralization, The Evolution of Cryptocurrencies, Open Consensus and Bitcoin, Smart Contracts and the Permissioned Models of Blockchain			
Unit – II	Blockchain Element and Consensus		6 Hours
Blockchain Elements, Permissionless Model and Open Consensus, Nakamoto Consensus (Proof of Work, Limitations of PoW: Forking and Security, Ethereum, State Machine Replication as Distributed Consensus, Paxos, Paxos Safety and Liveness,			
Unit – III	Byzantine Agreement Protocols and Hyperledger Fabric		5 Hours
Byzantine Faults, Safety and Liveness of PBFT, Enterprise Blockchains, Hyperledger Fabric, Consensus Scalability			
Unit – IV	Bitcoin and ByzCoin		5 Hours
Bitcoin-NG, Collective Signing (CoSi), Algorand, Identity Management			



Unit – V	Blockchain Interoperability and Hyperledger Aries	4 Hours
Blockchain Interoperability, Hyperledger Indy I , Hyperledger Indy II, Blockchain Security		
Unit – VI	A Potential Use Case From a Critics Perspective	4 Hours
Blockchain in Financial Services, Public Sector Use Cases, Blockchain for Decentralized Marketplace		
References		
1. https://onlinecourses.nptel.ac.in/noc26_cs34/preview 2. Imran Bashir, Mastering Blockchain: Distributed Ledger Technology, Decentralization, and Smart Contracts Explained, 2nd Edition https://users.cs.fiu.edu/~prabakar/cen5079/Common/textbooks/Mastering_Blockchain_2nd_Edition.pdf 3. Arvind Narayanan, Joseph Bonneau, Edward Felten, Andrew Miller and Steven Goldfeder, Bitcoin and Cryptocurrency Technologies: A Comprehensive Introduction, Princeton University Press (July 19, 2016). 4. Wattenhofer, The Science of the Blockchain Antonopoulos, Mastering Bitcoin: Unlocking Digital Cryptocurrencies 5. Satoshi Nakamoto, Bitcoin: A Peer-to-Peer Electronic Cash System		



WALCHAND INSTITUTE OF TECHNOLOGY, SOLAPUR
(An Autonomous Institute)
B.Tech. (Computer Science and Engineering), Semester-VIII

23CSU8EN23T - Programme Elective VI – QUANTUM COMPUTING

Teaching Scheme		Examination Scheme	
Lectures	2 Hours/week	ESE	50 Marks
Credits	2		
Introduction			
This course provides a foundational understanding of quantum computing concepts, architectures, and algorithms. It equips engineering students with the knowledge required to analyse and develop quantum solutions for complex computational problems.			
Course Prerequisite			
Students should have basic knowledge of linear algebra, probability theory, and classical algorithms. Familiarity with Python programming and fundamentals of computer architecture is recommended.			
Course Objectives:			
1. To introduce fundamental principles of quantum mechanics relevant to quantum computation. 2. To develop the ability to Analyze and design quantum circuits using quantum gates and registers. 3. To study quantum algorithms and understand their computational advantages over classical algorithms. 4. To provide hands-on exposure to quantum programming platforms and emerging real-world applications.			
Course Outcomes:			
At the end of course the student will be able to 1. Understand the basic principles of quantum computing and quantum mechanics. 2. Design and analyze simple quantum circuits and gates. 3. Implement basic quantum algorithms using quantum programming tools. 4. Identify applications and future challenges of quantum computing.			
Unit – I	Foundations of Quantum Mechanics		6 Hours
Limitations of classical computation and motivation for quantum computing, Basic postulates of quantum mechanics for computation, Qubits and Quantum States, Dirac Notation , Superposition and Normalization, Measurement and Probability, Bloch Sphere Representation			
Unit – II	Quantum Gates and Circuit Model		6 Hours
Quantum Registers, Single Qubit Gates: X, Y, Z, H, Phase, Multi-Qubit Gates: CNOT, Toffoli, SWAP Quantum Circuit Model, Reversible Computing, Quantum Entanglement, No-Cloning Theorem			
Unit – III	Quantum Algorithms		6 Hours
Classical vs Quantum Algorithmic Speedup, Deutsch–Jozsa Algorithm, Grover’s Search Algorithm, Shor’s Factoring Algorithm, Quantum Fourier Transform (QFT), Applications in Cryptography			



Unit – IV	Quantum Error Correction & Noise	6 Hours
Decoherence and Quantum Noise, Quantum Errors: Bit-Flip, Phase-Flip, Quantum Error Correction Codes, Shor Code and Steane Code, Fault-Tolerant Quantum Computing, Physical limitations of Quantum Systems.		
Unit – V	Quantum Programming & Platforms	6 Hours
Quantum Programming Concepts, Introduction to Qiskit / Cirq, Writing Quantum Programs, Quantum Simulators vs Real Quantum Hardware, Cloud-based Quantum Platforms, Case Studies: Optimization, AI, Chemistry, Future Scope and Industrial applications		
Text Books		
Quantum Computation and Quantum Information – Michael A. Nielsen, Isaac L. Chuang, 10th Anniversary Edition, Cambridge University Press. An Introduction to Quantum Computing – Phillip Kaye, Raymond Laflamme, Michele Mosca, First Edition, Oxford University Press.		
Reference Books :		
Quantum Computing: A Gentle Introduction – Eleanor Rieffel, Wolfgang Polak, First Edition, MIT Press. Qiskit Documentation – IBM Quantum Team, Qiskit Official Documentation, IBM.		
e-Resources		
UNIT I – Fundamentals of Quantum Mechanics https://www.geeksforgeeks.org/blogs/introduction-quantum-computing/ https://quantum.country/qcvc UNIT II – Quantum Gates and Circuits https://monitsharma.github.io/Learn-Quantum-Computing-with-Qiskit/ https://qiskit.org/textbook/ch-gates/what-is-a-quantum-gate.html https://quantumcomputinguk.org/gates-circuits-explained/ UNIT III – Quantum Algorithms https://quantum.cloud.ibm.com/learning/modules/computer-science/grovers?authuser= https://arxiv.org/abs/1903.04359 UNIT IV – Quantum Error Correction & Noise https://qosf.org/learn_quantum https://quantumcomputinguk.org/quantum-error-correction-explained Unit V – Quantum Programming & Platforms https://quantumai.google/cirq/start/basics https://learn.microsoft.com/en-us/azure/quantum/qsharp-overview?utm_source= https://quantumcountry.com/ https://arxiv.org/abs/2111.15352 https://qiskit.org/advocacy/applications		



WALCHAND INSTITUTE OF TECHNOLOGY, SOLAPUR
(An Autonomous Institute)
Final Year B.Tech. (Computer Science and Engineering), Semester-VIII

23CSU8PR4L:Project Phase-II

Teaching Scheme		Examination Scheme	
Lectures	--	ICA	100 Marks
Practical	4 Hours/week	POE	50 Marks
Credits	2		
Introduction:			
Project work is kept in the final year of engineering so that students' will apply their knowledge gained through previous classes to create and evaluate innovative things. In this it is expected to solve some pressing problem related to industry or society. While carrying out the work many qualities are developed in students such as problem-solving ability, modern tool usage, leadership, ethics, communication, project management, finance and lifelong learning etc.			
Course Prerequisite:			
Basic knowledge of all the core subjects completed during the previous semesters.			
Course Objectives:			
1. To familiarize with the industry landscape. 2. To gain comprehensive knowledge of software and hardware development tools and techniques for solving real-world problems. 3. To learn to demonstrate professional and ethical responsibilities.			
Course Outcomes:			
After completing the course, students will be able to 1. Analyze technological alternatives for developing IT solution with relevance to environment and sustainability. 2. Explore state-of-art tools and FOSS alternatives to develop solutions meeting societal and professional needs. 3. Develop a system through Software Development Life Cycle. 4. Demonstrate ability to engage in teamwork while observing professional ethics. 5. Write and present a well-organized project report. 6. Inculcate habit of self-study and lifelong learning.			
The group has to complete the project work and submit the hard bound copy of the report of the same at the end of this semester			

